Research methods



Plan for today

(1) Be part of our online lecture series
(2) Assessment: Research proposal
(3) Research ethics
(4) Using (and not misusing) research tools
(5) AI declarations
(6) Questions about your projects answered

Be part of our online lecture series

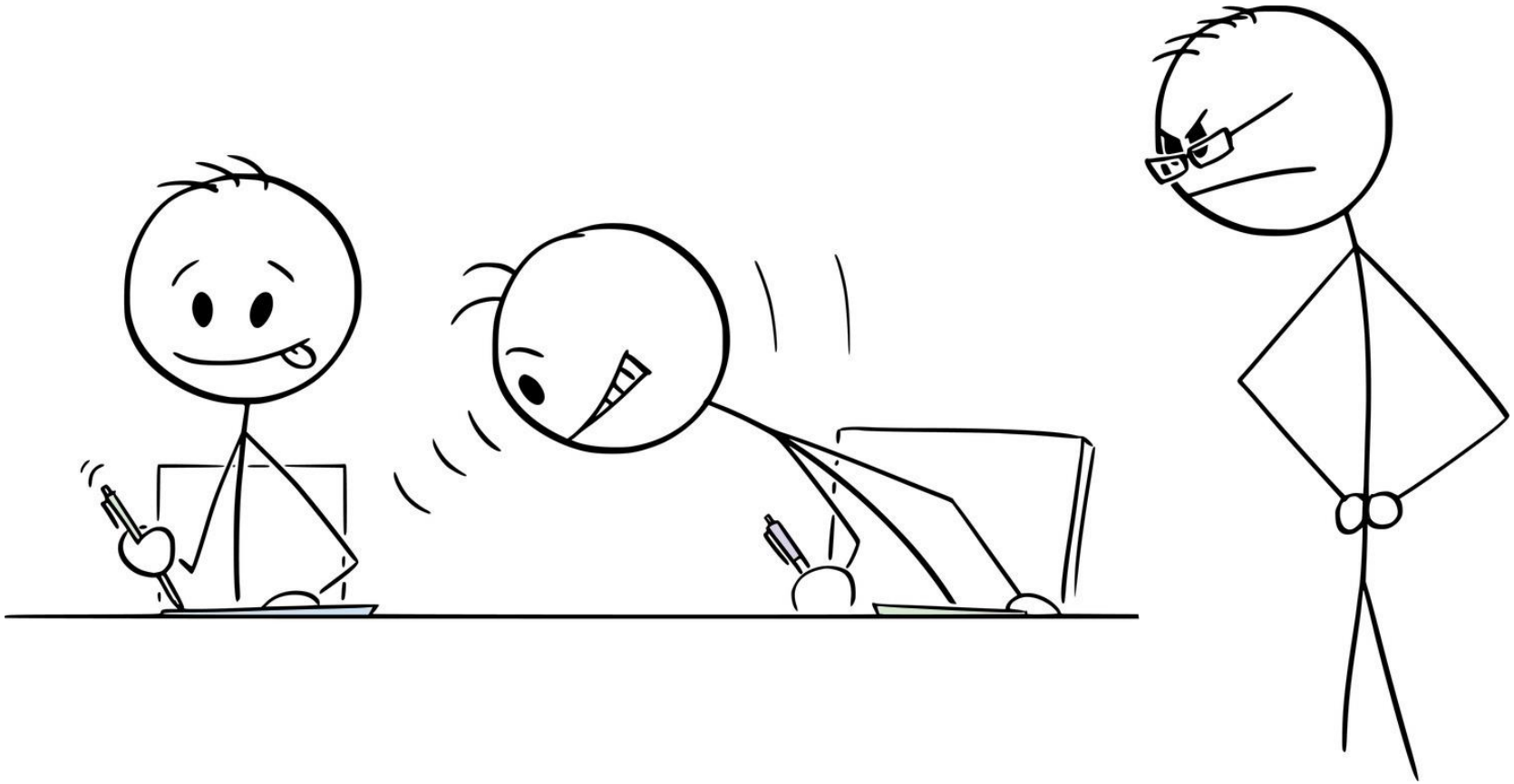
Our school is making an online lecture series about responsible AI.
If you'd like to be involved, you can

1) contribute your favourite online lecture series to our set of examples,
2) join the discussion about the content of the lectures,
3) apply to be a presenter of one or more of the lectures.

If you're interested, you can send me an email or Whatsapp on 0618595112.



Assessment: Research proposal



Assessment: Research proposal

The purpose of this assessment is for you to put forward a detailed and coherent description of your proposed project and to show that it is well-motivated and feasible.
The format of this assessment is a research proposal.

Assessment: Research proposal

The due date of the assessment is Sunday 17 May 23:59.

Assessment: Research proposal

For this assessment, you need to

- prepare a research proposal,
- make sure that your proposal is at least 4000 words,
- make sure that your proposal is typeset using LaTeX,
- make sure that you complete the AI declaration form and include it in the front matter of your proposal,
- make sure that each source match on the generated Turnitin report (other than the proposal template and the AI declaration form) is ≤1%,
- submit your proposal via the submission page (and not directly to your supervisor(s)),

Assessment: Research proposal

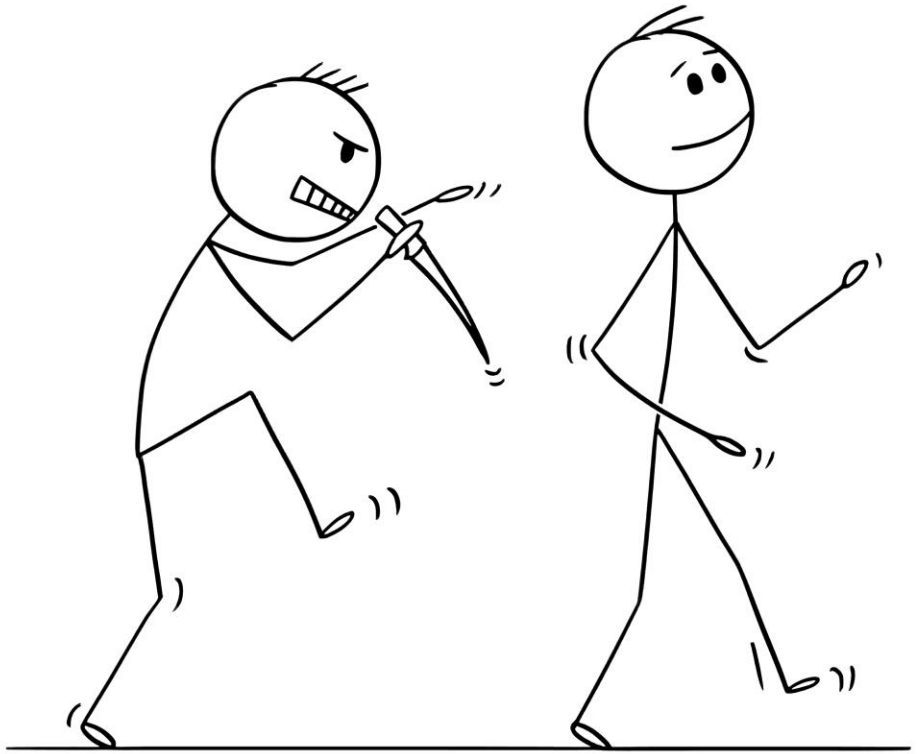
For this assessment, you need to

- schedule a brief proposal defence meeting with your supervisor(s) (for your supervisor(s) to make sure that you are sufficiently familiar with your proposed research and that they're not suspicious of your proposal being AI-generated in ways that it shouldn't be).

Assessment: Research proposal

(discussion of research proposal example on Overleaf)

Research ethics



Research ethics

If your project

(1) will use the data of humans (collected, secondary, or aggregate) or
(2) will use (begin with) AI-generated or synthetic data or
(3) might otherwise impact humans, other animals, or the environment,

you probably need to obtain ethics clearance for it.

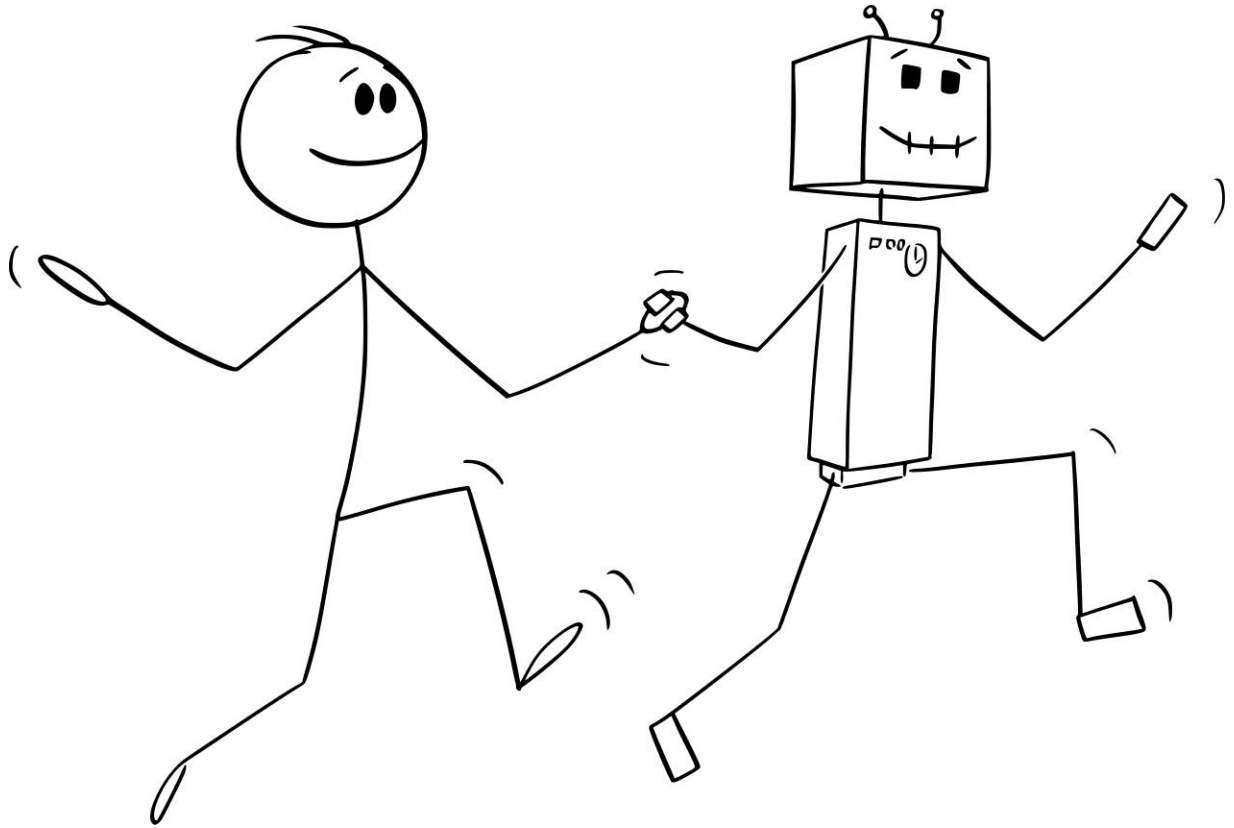
Research ethics

If your project requires ethics clearance and you don't obtain it before you begin using your data, your degree and/or any publications related to your project could be at risk.

Research ethics

For further details or to apply for ethics clearance, you can send me an email.

Using (and not misusing) research tools



Using (and not misusing) research tools



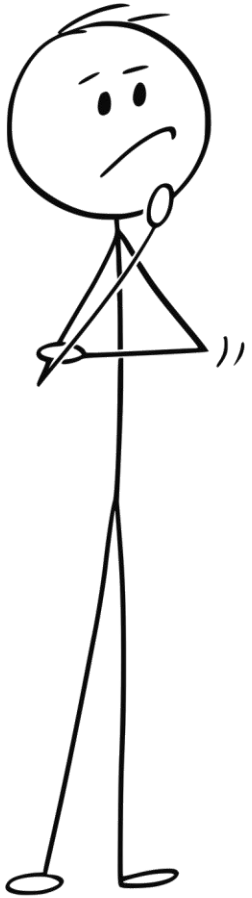
What counts as misusing a tool?

In all cases, you're misusing a tool if you're not honest about using it and about how you've used it.

For example, if you smuggle a calculator into a test in which they're not permitted and use it to work out some of the answers, this counts.

Or, for example, if you generate a summary of an article using an LLM and submit it as if you wrote it yourself, this counts.

Using (and not misusing) research tools



What should count as misusing a tool in general?

For example, if you generate an image to add to your poster or presentation slides, this doesn't seem like misuse. If a graphic design student generates an image to add to their portfolio, it does.

Or, for example, if you use Gemini Deep Research to formulate your research question, this seems like misuse. But if you use it to come up with some range of possible questions within a particular area, it doesn't.

Using (and not misusing) research tools



What should count as misusing a tool in general?

Fortunately, for our purposes, we don't need to find a principled way of distinguishing between these (and other possible) examples.

In all cases, you should check with your supervisor whether using a certain tool in a certain way is all right.

AI declarations

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☐
3. Methods and Experiment Design (experiment setup, model tuning ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ☐
8. Editing (grammar, readability) ☐
9. Writing (text generation, document structuring) ☐
10. Citation Formatting (structuring, organising) ☐

AI declarations

An AI declaration is a statement that indicates whether and how you've used any generative-AI tools in a given assessment or submission.

For your second and third assessments for our course, you need to include such a statement.

You can use the form posted on the course site or you can include the details at the beginning or end of the assessment itself.

Questions about your projects answered

